

Assignment No 08

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Question:

Design a base class Shape with two double-type data members to represent the dimensions of the shape. Include member functions to input the dimensions and a pure virtual function compute_area() for calculating the area. Accepts the dimensions of a triangle or rectangle from the user. Uses dynamic binding to call the appropriate compute_area() function. Displays the calculated area of the shape.

Code:

```
/*
Design a base class Shape with two double-type data members to represent the
dimensions of the shape. Include member functions to input the dimensions and a
pure virtual function compute_area() for calculating the area.
Derive two classes: Triangle and Rectangle from Shape and override the
compute_area() function to calculate the area according to the specific shape.
Write a program that:
a) Accepts the dimensions of a triangle or rectangle from the user.
b) Uses dynamic binding to call the appropriate compute_area() function.
c) Displays the calculated area of the shape.
*/

#include<iostream>
#include<iomanip>
#include<limits>
using namespace std;

class Shape {
protected:
    double dim1, dim2;

public:
    void inputDimensions(){
        while(true){
            cout << "Enter first dimension: ";
            cin >> dim1;
            if(!cin.fail() && dim1 > 0){
                break;
            }
            cout << "Invalid input! Enter a positive number.\n";
            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
        }

        while(true){
            cout << "Enter second dimension: ";
            cin >> dim2;
            if(!cin.fail() && dim2 > 0){
                break;
            }
            cout << "Invalid input! Enter a positive number.\n";
            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
        }
    }

    virtual double compute_area() = 0;        // pure virtual

    virtual void displayArea(){
        cout << "Area: " << compute_area() << endl;
    }
};
```

```

class Triangle : public Shape{
public:
    double compute_area() override{
        return 0.5 * dim1 * dim2;
    }
};

class Rectangle : public Shape{
public:
    double compute_area() override{
        return dim1 * dim2;
    }
};

int main(){
    int choice;
    Shape* shape;

    while(true){
        cout << "\nSelect Shape:\n";
        cout << "1. Triangle\n";
        cout << "2. Rectangle\n";
        cout << "0. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        if(cin.fail()){
            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');
            cout << "Invalid input!\n";
            continue;
        }

        switch(choice){
            case 1:
                shape = new Triangle();
                shape->inputDimensions();
                cout << "\n--- Triangle ---\n";
                shape->displayArea();
                break;

            case 2:
                shape = new Rectangle();
                shape->inputDimensions();
                cout << "\n--- Rectangle ---\n";
                shape->displayArea();
                break;

            case 0:
                cout << "Exiting Program\n";
                return 0;

            default:
                cout << "Invalid choice!\n";
        }
    }

    return 0;
}

```

Output:

```
PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 8> .\shape.exe

Select Shape:
1. Triangle
2. Rectangle
0. Exit
Enter choice: 1
Enter first dimension: 3
Enter second dimension: 2

--- Triangle ---
Area: 3

Select Shape:
1. Triangle
2. Rectangle
0. Exit
Enter choice: 2
Enter first dimension: 5
Enter second dimension: 3

--- Rectangle ---
Area: 15

Select Shape:
1. Triangle
2. Rectangle
0. Exit
Enter choice: 0
Exiting Program
PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 8> |
```